



Data Quality Assessment Report

Customer List.xls

Sheet: Sheet1

55.8

Overall Quality Score / 100

Fair

Processing Date: 2026-08-05 01:46:57

Engine Version: Phase 1

Execution Time: 0.68s

Generated By: Data Quality Engine (Phase 1)

Executive Summary

Dataset:	Customer List.xls
Rows / Columns:	533 rows x 30 columns
Checks Executed:	8
Overall Quality Score:	55.8 / 100 (Fair)
Dimensions Excluded:	freshness

Critical Findings

- Missing Values: 19 column(s) affected, 4357 issue(s) found
- Duplicates: 1 column(s) affected, 5 issue(s) found
- Validity: 3 column(s) affected, 542 issue(s) found
- PII detected in 12 column(s) -- masking required before sharing

Positive Findings

- Type Mismatch: no issues found
- Schema Quality: no issues found
- Freshness: no issues found

Data Readiness

Ready with Moderate Cleaning

Business Impact & Risk Summary

Missing Values

Critical

Business Impact: May produce inaccurate analytics and unreliable reporting.

Duplicates

Critical

Business Impact: Can create duplicate customers/orders and inflate revenue or count metrics.

Outliers

Medium

Business Impact: May distort averages, totals, and any ML model trained on this data.

Consistency

Medium

Business Impact: Grouping and aggregation become unreliable (e.g. 'Paid' vs 'paid' split apart).

Validity

High

Business Impact: Values that look present may still be unusable or logically wrong.

PII Exposure

Critical

Business Impact: Privacy and compliance risk if this data is shared unmasked.

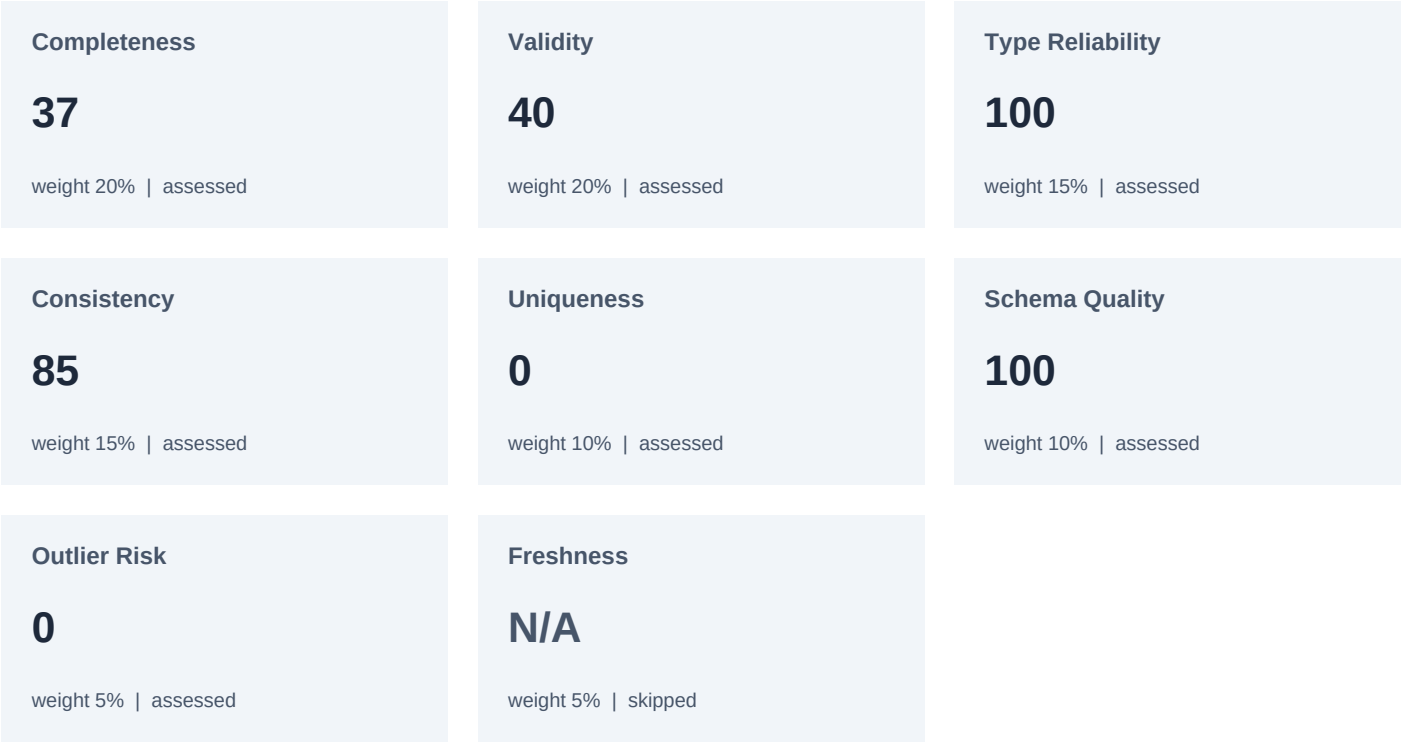
Dataset Overview

Rows:	533
Columns:	30
Header Row:	3
Processing Time:	0.68s

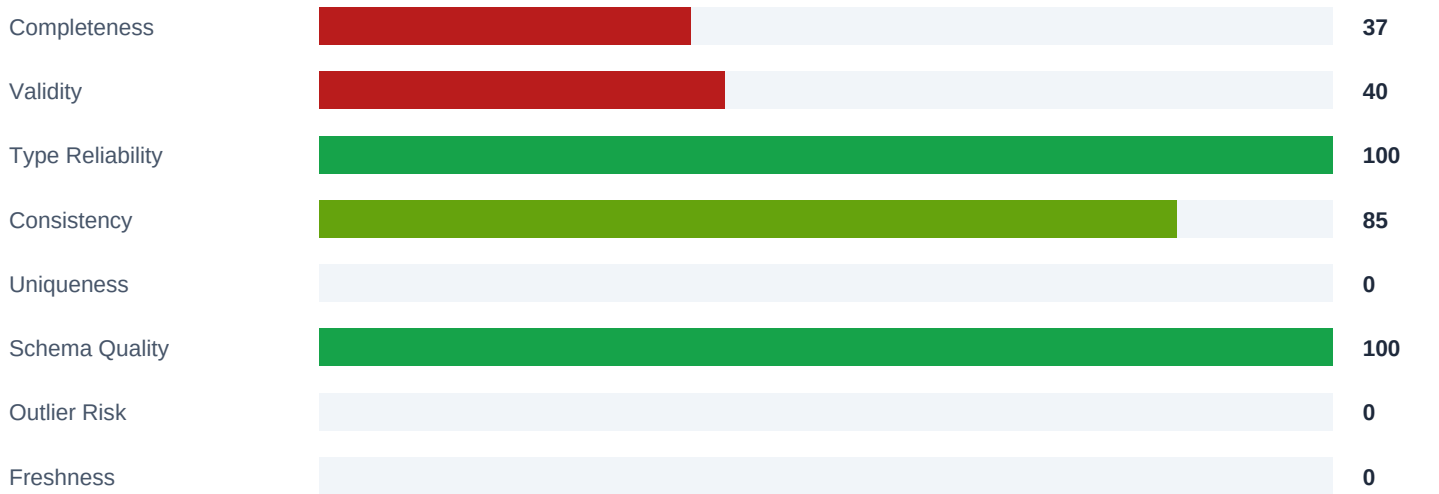
Column Classification

Measurement Columns:	2
Identifier Columns:	7
PII Columns:	9
Free-Text Columns:	6
Categorical Columns:	6
Date Columns:	0

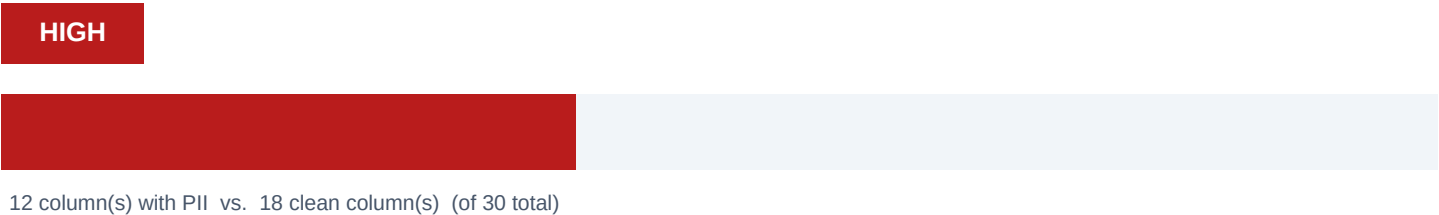
Data Quality Dashboard



Dimension Scores at a Glance



Privacy Risk (separate from score above)



Types found: ADDRESS, CARD, CNIC, EMAIL, MOBILE, PHONE

Missing Values

Critical

Columns Checked:	30
Columns with Issues:	19
Total Issues Found:	4357
Affected Columns:	Address Line 1, Address Line 2, Address Line 3, City, Company Reg No., Customer EORI, Customer VAT Number, Email Addresses Generic, Fax, Mobile, Other, POs/Sales Quotes, Post Code, Remittances, Order Acks & Shipments, Sales Invoices/Credits

Business Impact

May produce inaccurate analytics and unreliable reporting.

Recommendation

Improve data collection at source; consider a required-field rule for critical columns.

Sample Findings

- Address Line 1: 9 issue(s)
- Address Line 2: 154 issue(s)
- Address Line 3: 351 issue(s)
- City: 31 issue(s)
- Post Code: 69 issue(s)

Duplicates

Critical

Columns Checked:	1
Columns with Issues:	1
Total Issues Found:	5

Business Impact

Can create duplicate customers/orders and inflate revenue or count metrics.

Recommendation

Add a uniqueness constraint on the business key at the source system.

Sample Findings

- None: 5 issue(s) note=row_indices are 0-based positions in the loaded dataframe (after header confirmation), not Excel row numbers.

Type Mismatch

None

Columns Checked:	30
Columns with Issues:	0
Total Issues Found:	0

Business Impact

Charting, grouping, and trend analysis may silently break or mislead.

Recommendation

Standardize the column's expected type at data entry.

Outliers

Medium

Columns Checked:	13
Columns with Issues:	2
Total Issues Found:	197
Affected Columns:	Credit Limit, Overall Credit Limit

Business Impact

May distort averages, totals, and any ML model trained on this data.

Recommendation

Manually review flagged values before using this column in analysis or ML.

Sample Findings

- Credit Limit: 98 issue(s) note=high concentration on one value -- IQR bounds may be unreliable for this distribution
- Overall Credit Limit: 99 issue(s) note=high concentration on one value -- IQR bounds may be unreliable for this distribution

Schema Quality

None

Columns Checked:	30
Columns with Issues:	0
Total Issues Found:	0

Business Impact

Downstream tools and analysts may misread or skip unclear columns.

Recommendation

Rename unclear/duplicate/blank columns before this file is reused.

Consistency

Medium

Columns Checked:	30
Columns with Issues:	2
Total Issues Found:	4
Affected Columns:	Post Code, Sales Invoices/Credits

Business Impact

Grouping and aggregation become unreliable (e.g. 'Paid' vs 'paid' split apart).

Recommendation

Standardize values at entry (dropdowns/lookups instead of free text).

Sample Findings

- Post Code: 1 issue(s) examples=[{'normalized': 'rg19 4jz', 'canonical': 'RG19 4JZ', 'variants': {'RG19 4JZ': 3, 'RG19 4Jz': 1}}]
- Sales Invoices/Credits: 3 issue(s) examples=[{'normalized': 'gb-invoices-210@schneider-electric.com', 'canonical': 'GB-Invoices-210@schneider-electric.com', 'variants': {'GB-Invoices-210@schneider-electric.com': 1, 'GB-Invoices-210@schneider-ele

Validity

High

Columns Checked:	8
Columns with Issues:	3
Total Issues Found:	542
Affected Columns:	Custom Hold Status, Customer No.,Add. Code, Email Addresses Generic

Business Impact

Values that look present may still be unusable or logically wrong.

Recommendation

Fix values that fail format/logic rules at the source before further use.

Sample Findings

- Customer No.,Add. Code: 5 issue(s)
- Email Addresses Generic: 4 issue(s)
- Custom Hold Status: 533 issue(s)

Freshness

None

Columns Checked:	30
Columns with Issues:	0
Total Issues Found:	0

Business Impact

Decisions may be based on stale data without anyone noticing.

Recommendation

Confirm this data is refreshed on the expected schedule.

Column Quality Matrix

Column	Role	Score	Issues	Severity	Recommendation
Sales Invoices/Credits	identifier	60	121	High	Standardize values at entry (dropdowns/lookups instead
Post Code	identifier	67	70	High	Standardize values at entry (dropdowns/lookups instead
Email Addresses Generic	pii	71	150	High	Review flagged values.
Address Line 1	pii	80	9	High	Improve data collection at source; consider a required-
Address Line 2	pii	80	154	High	Improve data collection at source; consider a required-
City	free_text	80	31	High	Improve data collection at source; consider a required-
POs/Sales Quotes	free_text	80	219	High	Improve data collection at source; consider a required-
Remittances, Order Acks & Sh	free_text	80	223	High	Improve data collection at source; consider a required-
Statements	free_text	80	124	High	Improve data collection at source; consider a required-
Ship Address VAT Number	identifier	80	465	High	Improve data collection at source; consider a required-
Address Line 3	pii	83	351	Medium	Improve data collection at source; consider a required-
Telephones Landline	pii	83	171	Medium	Improve data collection at source; consider a required-
Fax	pii	83	439	Medium	Improve data collection at source; consider a required-
Toll Free	pii	83	488	Medium	Improve data collection at source; consider a required-
Other	pii	83	447	Medium	Improve data collection at source; consider a required-
Mobile	pii	83	441	Medium	Improve data collection at source; consider a required-
Customer VAT Number	identifier	83	160	Medium	Improve data collection at source; consider a required-
Customer EORI	free_text	83	156	Medium	Improve data collection at source; consider a required-
Company Reg No.	identifier	83	146	Medium	Improve data collection at source; consider a required-
Credit Limit	measurement	83	98	Medium	Manually review flagged values before using this column
Overall Credit Limit	measurement	83	99	Medium	Manually review flagged values before using this column
Custom Hold Status	categorical	83	533	Medium	Review flagged values.
Customer No.	identifier	100	0	None	No action needed.
Company Name	free_text	100	0	None	No action needed.
Default?	categorical	100	0	None	No action needed.
Add. Code	identifier	100	0	None	No action needed.
Country	categorical	100	0	None	No action needed.
External Rep	categorical	100	0	None	No action needed.
Internal Rep	categorical	100	0	None	No action needed.
Pay Terms	categorical	100	0	None	No action needed.

Top Critical Issues

1. Missing Values -- Address Line 1

Critical

Impact: May produce inaccurate analytics and unreliable reporting.

Action: Improve data collection at source; consider a required-field rule for critical columns.

2. Missing Values -- Address Line 2

Critical

Impact: May produce inaccurate analytics and unreliable reporting.

Action: Improve data collection at source; consider a required-field rule for critical columns.

3. Missing Values -- Address Line 3

Critical

Impact: May produce inaccurate analytics and unreliable reporting.

Action: Improve data collection at source; consider a required-field rule for critical columns.

4. Missing Values -- City

Critical

Impact: May produce inaccurate analytics and unreliable reporting.

Action: Improve data collection at source; consider a required-field rule for critical columns.

5. Missing Values -- Company Reg No.

Critical

Impact: May produce inaccurate analytics and unreliable reporting.

Action: Improve data collection at source; consider a required-field rule for critical columns.

6. Missing Values -- Customer EORI

Critical

Impact: May produce inaccurate analytics and unreliable reporting.

Action: Improve data collection at source; consider a required-field rule for critical columns.

7. Missing Values -- Customer VAT Number

Critical

Impact: May produce inaccurate analytics and unreliable reporting.

Action: Improve data collection at source; consider a required-field rule for critical columns.

8. Missing Values -- Email Addresses Generic

Critical

Impact: May produce inaccurate analytics and unreliable reporting.

Action: Improve data collection at source; consider a required-field rule for critical columns.

9. Missing Values -- Fax

Critical

Impact: May produce inaccurate analytics and unreliable reporting.

Action: Improve data collection at source; consider a required-field rule for critical columns.

10. Missing Values -- Mobile**Critical**

Impact: May produce inaccurate analytics and unreliable reporting.

Action: Improve data collection at source; consider a required-field rule for critical columns.

Appendix

Execution Time: 0.68s
Engine Version: Phase 1

Scoring Formula

Each dimension score = $100 \times (\text{passed checks} / \text{total non-error checks})$ for that dimension. Composite score = weighted average across scorable dimensions only (weights re-normalized when a dimension has no results). Privacy Risk is calculated and reported separately and is never subtracted from the composite score.

Quality Dimensions

- Completeness (weight: 20%)
- Validity (weight: 20%)
- Type Reliability (weight: 15%)
- Consistency (weight: 15%)
- Uniqueness (weight: 10%)
- Schema Quality (weight: 10%)
- Outlier Risk (weight: 5%)
- Freshness (weight: 5%)